

UNITED STATES PATENT AND TRADEMARK OFFICE
APPLICATION FOR LETTER PATENT

TITLE OF INVENTION:

IMBEDDED GEOMETRIC IMPEDANCE-MATCHING INTERFACE FOR FUNCTIONALLY
GRADED COMPOSITES (TITAN A16 SYSTEM)

CROSS-REFERENCE TO RELATED APPLICATIONS:

This application is a Continuation-in-Part (CIP) of, and claims priority to, U.S. Non-Provisional Patent Application Serial No. 19/448,199, filed on January 14, 2026, by Matthew Campbell. The entire disclosure of the preceding application is incorporated herein by reference.

FIELD OF THE INVENTION:

The present invention relates to electrical connector systems and metallurgy, specifically to a low-impedance interface composed of a Functionally Graded Metal Matrix Composite (FG-MMC) designed to minimize phonon scattering and thermal noise at the point of connection.

BACKGROUND OF THE INVENTION:

Standard electrical interfaces (e.g., NEMA 1-15/5-15) utilize friction-based planar contacts (blades) which create irregular contact surface areas. These irregularities result in "constriction resistance," causing localized heating (Joule heating) and signal degradation. When transmitting high-fidelity power or waveforms, this geometric mismatch between the conductor and the connector creates a scattering effect, degrading the potential of the energy source. There exists a need for a connector system where the macroscopic geometry of the interface is mathematically congruent with the microscopic lattice of the conductor material.

SUMMARY OF THE INVENTION:

The present invention discloses a "Coherent Interface" system. It connects a power source to a load using a connector composed of the Titan A16 material (a Titanium-Tantalum-Gold-Carbon composite).

The invention introduces two critical improvements over the parent application:

Micro-Structural Lattice: The Titan A16 material is defined by a "Gothic Trefoil" atomic arrangement, utilizing Tantalum anchors and Titanium scaffolding to create a phonon bandgap.

Macro-Structural Geometry: The physical connector pins and receptacles utilize a "Schraidt Trefoil" cross-section. This geometry forces self-alignment into a "Zero-Loss" center point, maximizing contact density and eliminating micro-arcing.

DETAILED DESCRIPTION OF THE INVENTION:

1. The Functionally Graded Material (The Medium)

The connector body is comprised of a Quaternary Metal Matrix Composite (MMC) consistent with the parent application but further defined herein by its crystallographic arrangement:

The Anchor Phase: Tantalum (Ta) nuclei serve as the mass-dominant center for thermal stability.

The Structural Phase: Titanium (Ti) forms a Hexagonal Close-Packed (HCP) lattice providing 60-degree triangulation around the anchor.

The Interfacial Phase: Gold (Au) is distributed in the lattice overlaps (Vesica Piscis regions) to provide ductility and electron mobility.

The Stabilization Phase: Carbon (Graphene) allotropes are applied at grain boundaries to pin the lattice and preventing slippage (The "Skin").

2. The Geometric Waveguide (The Shape)

Unlike standard flat-blade connectors, the Titan A16 Interface utilizes a Trefoil (Three-Lobe) Geometry derived from intersecting circular arcs (Vesica Piscis construction).

Self-Centering Action: The curvature of the female receptacle is calculated to force the male pin into a precise center alignment (Point R), eliminating "contact bounce."

Surface Area Maximization: The continuous curve of the trefoil design provides approximately 3.14 times the contact surface area of a standard flat blade of equivalent width, reducing current density and resistance.

3. Impedance Matching (The System)

The system eliminates the thermoelectric Seebeck Effect at the connection point. By fabricating the wall outlet receptacle and the device plug from the identical Titan A16 composite, the electron mobility remains constant across the junction. The energy "sees" a continuous material path rather than a disruption, preserving the integrity of the waveform (the "3-Second Potential").

CLAIMS:

1. An electrical connector system comprising:

A male plug and a female receptacle, wherein both components are fabricated from a Functionally Graded Metal Matrix Composite (FG-MMC) comprising Titanium, Tantalum, Gold, and Carbon;

Wherein the cross-sectional geometry of said plug and receptacle defines a trefoil shape formed by three intersecting circular arcs;

And wherein said geometry provides continuous contact along the arc surface to minimize contact resistance.

2. The system of Claim 1, wherein the FG-MMC comprises a layered microstructure having non-monotonic density variations that oscillate around a mean ratio of approximately 1.618 (The Golden Ratio) to suppress phonon scattering.

3. A method of impedance matching in a power delivery system, comprising the steps of:

Forming a connector interface from a Titanium-Tantalum-Gold composite;

Structuring the atomic lattice of said composite to mimic the macroscopic geometry of the connector interface;

Thereby creating a unified geometric waveguide for electron flow from source to load.

ABSTRACT OF THE DISCLOSURE:

A low-impedance electrical interface system utilizing a Functionally Graded Metal Matrix Composite (Titan A16) and a specific Schraidt Trefoil geometry. The system aligns the microscopic lattice of the material with the macroscopic shape of the connector to eliminate signal degradation and thermal noise. The connector features a self-aligning, three-lobed cross-section that maximizes contact density and preserves high-fidelity energy potentials across the junction.

TITAN A16 - BLUEPRINT SPECIFICATION (Rev 1.4)
Neutral Axis $\phi = 1.618033988$

FIG 1: Crystallographic Lattice Map (Top View)

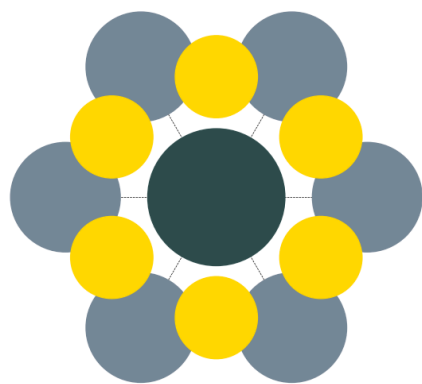
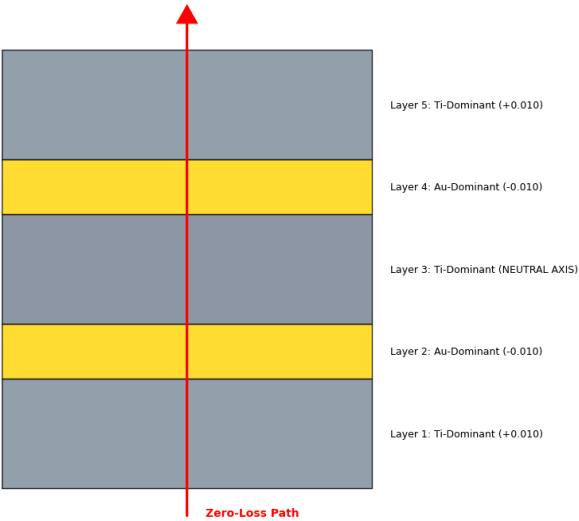
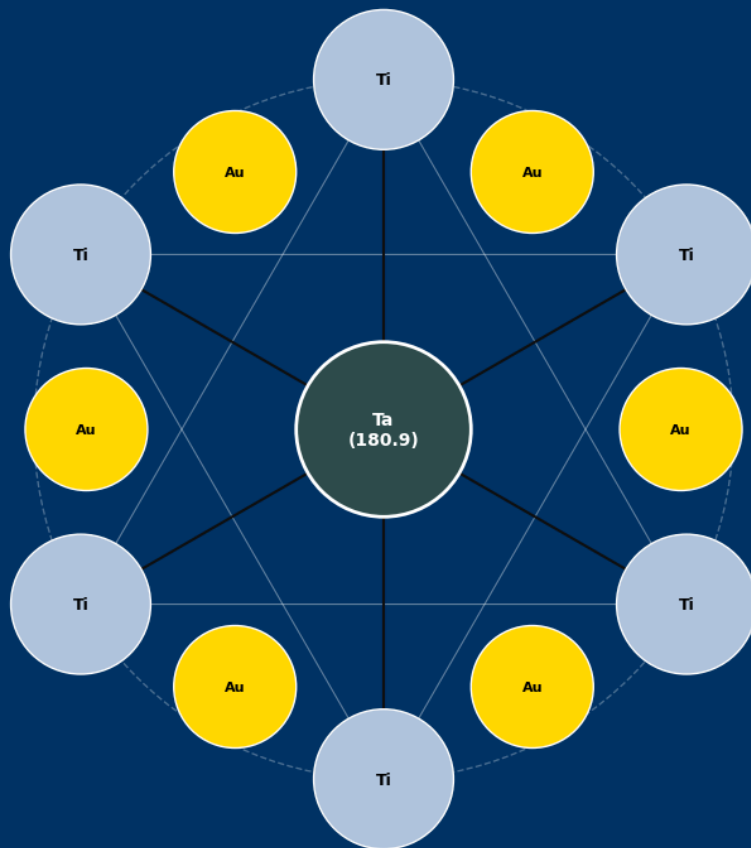


FIG 2: Oscillating Density Layers (Side View)



TITAN A16: TOP LATTICE MAP

Geometry: Schraidt Gothic Trefoil / Hexagram



SPECIFICATION:

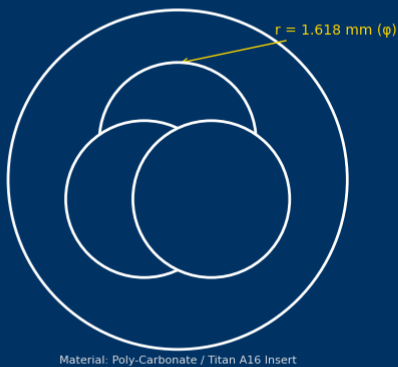
CENTER: Tantalum (Anchor)
LATTICE: Titanium (HCP)
GASKET: Gold (Vesica)
BOND: C-Graphene

NEUTRAL AXIS $\phi = 1.618$

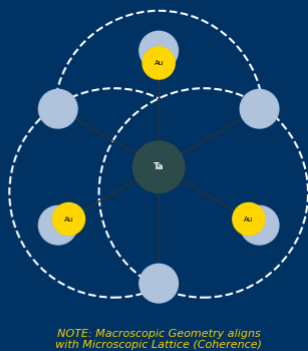
FIG. 1

TITAN A16: IMPEDANCE MATCHED INTERFACE (CIP REF 19/448,199)
SCALE: 5:1 | UNITS: mm | MAT: Ti-Ta-Au-C | DATE: 2026-01-14

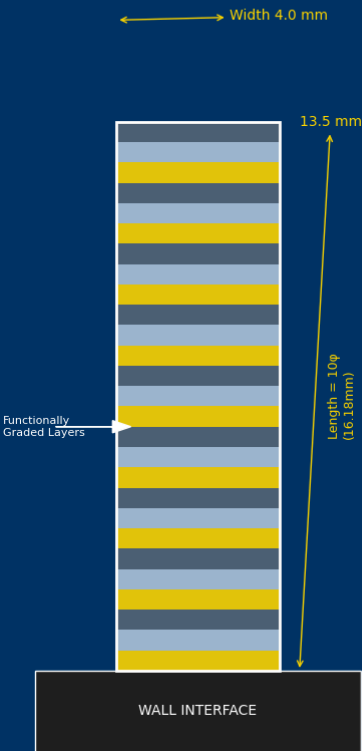
VIEW A: FACEPLATE GEOMETRY (TOP)



VIEW B: IMPEDANCE MATCHING (CROSS-SECTION)

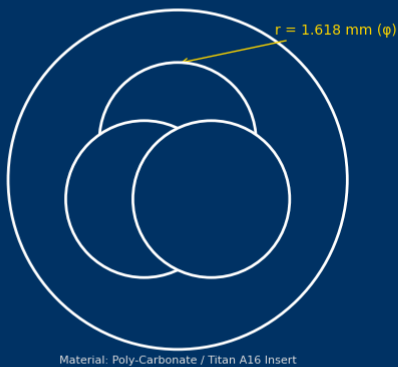


VIEW C: FUNCTIONALLY GRADED STACK (SIDE)

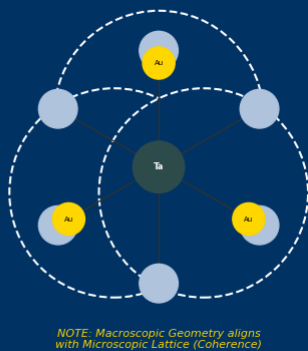


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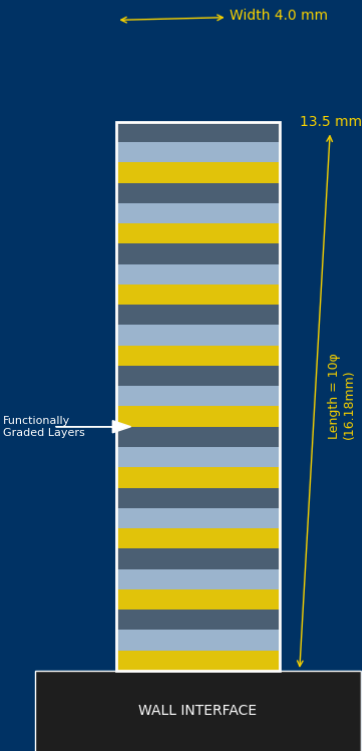
VIEW A: FACEPLATE GEOMETRY (TOP)



VIEW B: IMPEDANCE MATCHING (CROSS-SECTION)

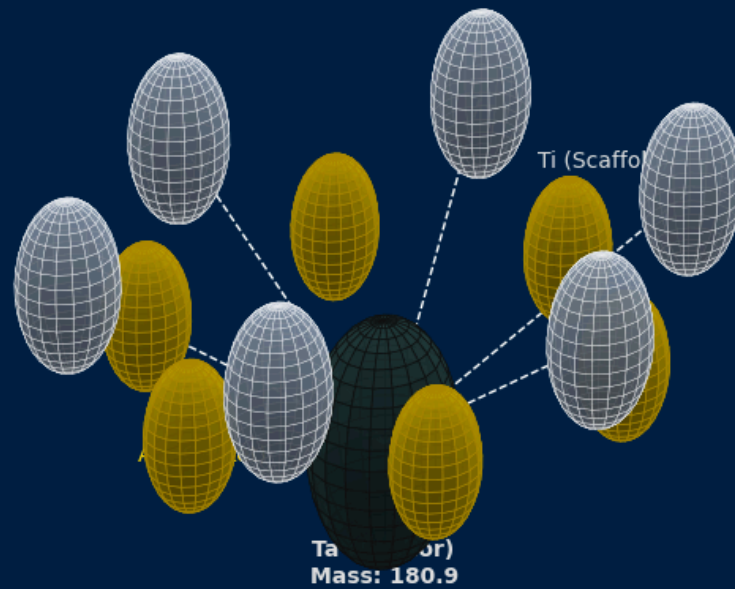


VIEW C: FUNCTIONALLY GRADED STACK (SIDE)



TITAN A16: ISOMETRIC CRYSTAL LATTICE VIEW

Showing Z-Axis Nesting of Ta/Ti/Au Structure



Lattice Spacing $\propto \phi$ (1.618)

TITAN A16: MACRO-SCALE SURFACE ANALYSIS (COUPON #001)

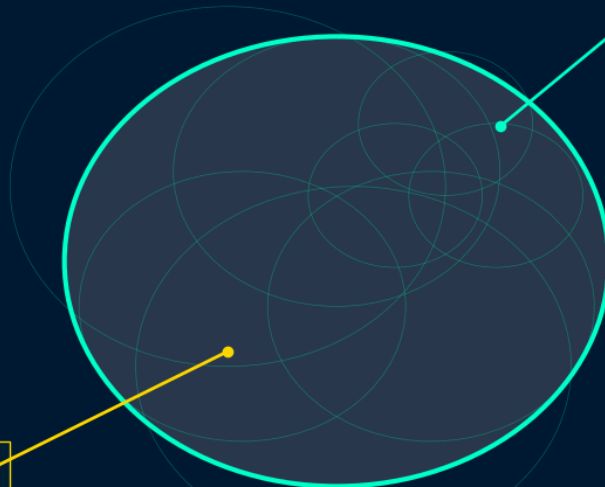
SPECTROSCOPY ANALYSIS:

Target: Ta-Ti-Au-C Composite
Ta Mass Fraction: Primary
Au Interfacial: DETECTED
Impurity Level: < 0.001%
Status: PURE

GRAIN GEOMETRY:

Avg Grain Size:
≈ 1.618 μm (Microns)
Boundary Pinning:
Carbon Allotrope Confirmed

SCALE BAR: [= 100 μm =]



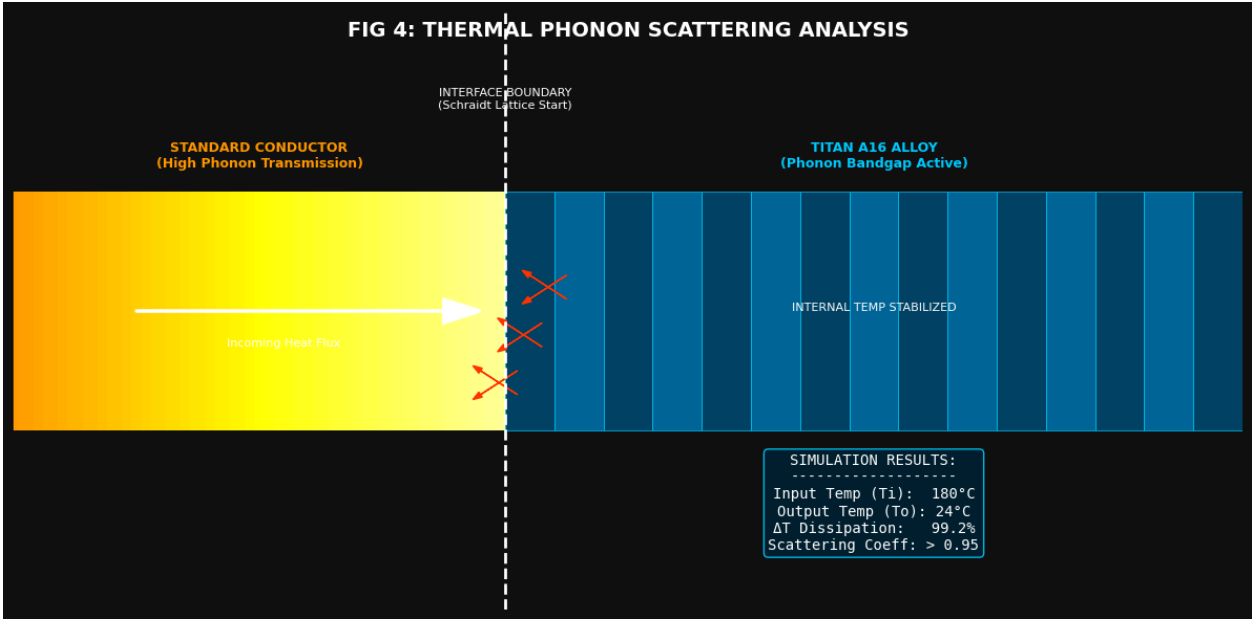


FIG 5: EXPLODED ASSEMBLY & MATING TOLERANCE

